Integrated geometric control of shell-volume response

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Abstract

A derivative-supremum condition can exclude concentrated smooth initial data even when their integrated geometric size remains bounded. We derive a direct H^1-to-observable estimate for the retained five-channel Einstein–scalar shell mean problem. On a physically supplied initial Cauchy section, the profile D is the relative difference of second scalar-level three-area variations, with fixed first perturbation and background scalar-label calibration. A tilted wave energy with a positive boundary value term controls the scalar history despite a mixed Dirichlet and nonselfadjoint Robin boundary block. The complete initial map gives $E(0) \le R^2/(5\times 10^6)$, and the boundary history obeys $|m| \le R/300$, where $R = \|D\|_{H^1([1,5/4])}$. The specified closed class $R \le 2\times 10^{31}$ retains strictly negative volume and expansion second coefficients at one proper time. An explicit sequence in the actual compact preparation image has bounded H^1 norm and unbounded old derivative-supremum norm, so this criterion covers data beyond that earlier sufficient criterion. It does not imply containment of the two budget balls or response instability. All coefficient, exact-family and linear-development premises remain explicit and conditional. The estimating energy is not an identified physical energy or preparation law; neither a physical class selection nor a uniform nonlinear parameter interval follows.

1 Model, fixed data, and the new estimate

The sign of a second response depends on both the common quadratic first source and the homogeneous second preparation. The retained shell model admits an explicit pair with opposite volume and expansion coefficient signs, at the same anchored proper time, while its first packet is fixed [6]. A subsequent geometric class, based on the supremum of an initial scalar-level area profile and its derivative, certifies negative signs throughout a specified budget [7]. The question here is whether integrated control of that same profile can give a direct response bound, even when the earlier derivative-supremum route fails.

The bulk normalization and one-copy outward shell equations are

$$\begin{aligned} S_{\mathrm{bulk}} &= \int (R_g/2 - |\mathrm{d}\phi|_g^2/2 - \mathcal{V}(\phi))\mathrm{d}\mu_g, \\ \mathrm{Ric}_{AB} &= \phi_A\phi_B + \tfrac{2}{3}\mathcal{V}g_{AB}, & \Box_g\phi &= \mathcal{V}', \\ K^{\mathrm{out}}_{ab} &= \tfrac{1}{6}\sigma(\phi)\gamma_{ab}, & n_{\mathrm{out}}\phi &= -\tfrac{1}{2}\sigma'(\phi). \end{aligned} \tag{1}$$

Two identical interior copies use their respective outward normals. Fix

$$W = 1-\phi+\phi^3/3, \qquad \mathcal{V} = (W')^2/2 - 2W^2/3, \qquad \sigma = 2W + 10^{-3}(1+c_\sigma\phi),$$
$$c_\sigma = 96714321425692/161839258930731. \tag{2}$$

The selected regular background and coordinates are

$$g_0 = \mathrm{d}y^2 + \rho(y)^2[-\mathrm{d}t^2 + \cosh^2 t\,\kappa], \qquad a = \rho_y/\rho > 0, \quad p = \phi_{0,y} \neq 0, \quad \mathrm{d}y = -\ell\rho\,\mathrm{d}r, \quad x = t/\ell, \tag{3}$$

where κ is the unit round three-sphere metric, $\ell = 10^{-6}$, the packet is $1 \le r \le 2$, and the exact selected shell radius is ρ_b. The target is

$$T = 5/4, \qquad \Delta = T - 1 = 1/4, \qquad \tau_* = \rho_b \ell T. \tag{4}$$

The exact ρ_b fixes the clock; its enclosure is not a replacement value. All numerical quantities below are in the model's fixed parameter and coefficient normalization, with no SI calibration.

We compare the same background, first canonical quadrupole seed, complete quadratic constraint and evolution sources, initial shell collar, canonical momentum and both scalar canonical second sectors, and second transverse-traceless initial pair. Parameter derivatives use

$$x_\varepsilon = x_0 + \varepsilon v + \tfrac{1}{2}\varepsilon^2 w + o(\varepsilon^2). \tag{5}$$

The common smooth flat seed has $Y = n_1^2 - n_2^2$, $\langle Y\rangle = 0$, $\langle Y^2\rangle = 1/6$ and $\Delta_\kappa Y = -8Y$; its exact normalization and the fixed base preparation are those specified in [6, 7]. The retained base response includes the complete first-field bulk and shell forcing, clock corrections and measure corrections.

We assume the selected smooth background and its authenticated continuous boxes; the equivariant weighted constraint inverse and nonlinear implicit-function hypotheses; the conditional exact-family fixed-shell development, subsidiary and boundary uniqueness; and identification of the common first derivative. Each fixed smooth preparation has C^2 parameter dependence in the retained lower C^4 topology on a compact neighborhood. An anchored intrinsic normal tube reaches the target after a possibly family-dependent shrinkage in ε. The exact five-mean projection, homogeneous difference equation, boundary rows and continuous operator caps in Section 3 are retained conditional premises. In particular the numerical caps are declared by the earlier `EXPLICIT_PREMISES.json` record [9]; they are not reproved by the new arithmetic. The new result is an a priori estimate for this stipulated coupled problem, not a new nonlinear existence theorem.

2 Geometric profile and complete initial map

On the supplied physical initial section Σ, let h be smooth, compact in the packet, flat at both endpoints, and satisfy

$$\int_{y_L}^{y_U} \rho^2 h\,\mathrm{d}y = 0, \qquad \chi(y) = \int_{y_L}^{y} \rho^2 h\,\mathrm{d}s, \qquad h = \rho^{-2}\chi_y, \quad y_L = y(2), \quad y_U = y(1). \tag{6}$$

The complete added raw canonical second datum is

$$(\Delta\gamma_2, \Delta\pi_2, \Delta\phi_2, \Delta\varpi_2) = \big(h\gamma_0 + \mathcal{L}_{\zeta\partial_y}\gamma_0, 0, 0, 0\big), \qquad \zeta = \frac{3(h_y - ah)}{2p^2} + \frac{9\chi}{2\rho^4 p^2}. \tag{7}$$

Its exact homogeneous constraint and zero-moment construction, and its conditional exact completion with the same full correction second derivative, are retained from [6]. No quadratic source is changed by this addition.

For $\varphi = \phi_0(y)$, use the scalar-level component in the annular collar that continues the background sphere. Its three-area is $A_\varepsilon(\varphi)$ and $A_0(\phi_0(y)) = 2\pi^2\rho^3$. At fixed scalar label define

$$D = \frac{\partial_\varepsilon^2 A_\varepsilon^h|_0 - \partial_\varepsilon^2 A_\varepsilon^{\mathrm{base}}|_0}{A_0}, \qquad D_r = (-\ell\rho p)\partial_\varphi D. \tag{8}$$

This derivative uses the fixed background scalar-label map, not an arbitrary coordinate of the perturbed metric. In a slightly enlarged collar, $p \neq 0$ gives regular continued level graphs. The retained C^2-into-C^4 dependence gives C^2-into-C^1 area dependence after the two spatial orders of loss described in [7]. Thus D and D_r exist. There is no assumption of exact rotational symmetry of the perturbed data.

The two first level displacements agree. Their second displacement difference is $-\Delta\phi_2/p$, and the common first-field determinant quadratics cancel after subtraction. For the angular metric mean J and scalar mean w this gives

$$D = \tfrac{3}{2}J - 3aw/p. \tag{9}$$

The raw representative has $J = h + 2a\zeta, w = 0$; after the pure second conditioning gauge it has $J = h, w = -p\zeta$. Both give the same D. This is the standard metric–scalar displacement cancellation on the given section. Simultaneous diffeomorphisms transport the section, fields and level component; choosing a different physical section is a different observable.

Set $Z = \chi/(\ell\rho^3)$, $b = \ell\rho a$, $g = \ell\rho p^2/(3a) > 0$, and $\beta = p/(2a)$. Then

$$Z_r = 3bZ - h, \qquad h_r = 3\ell^2 Z + (-b+g)h - \tfrac{2}{3}gD, \qquad (Z,h)(1) = (0,0). \tag{10}$$

This retained reconstruction determines the restricted preparation on $[1,T]$ from its profile there; whole-packet injectivity needs the full profile on $[1,2]$. It is not injectivity on unrestricted Cauchy data or surjectivity onto arbitrary profiles.

In (K, L, N, J, w) order the complete conditioned difference initial map is

$$\begin{aligned} &U_0 = (h,0,0,h,w), \qquad V_0 := U_{0,x} = (0,V_L,0,0,0), \qquad P_0 := U_{0,r} = (h_r,0,0,h_r,w_r), \\ &w = \beta(h - 2D/3), \qquad w_r = \beta_r(h-2D/3) + \beta(h_r - 2D_r/3), \qquad V_L = h_r + bh. \end{aligned} \tag{11}$$

In particular the mixed metric velocity and derivative of the scalar coefficient are included. Define the integrated profile size

$$R = \|D\|_{H^1(I)} := \left(\int_I (D^2 + D_r^2)\,\mathrm{d}r\right)^{1/2}, \qquad I = [1,T]. \tag{12}$$

It applies to the complete increment: $R(\lambda h) = |\lambda| R(h)$.

3 Tilted five-channel energy and boundary history

Write $|U|_{\mathsf{H}}^2 = U^{\mathsf{T}}\mathsf{H}U$ for $\mathsf{H} = \mathrm{diag}(1,2,1,3,1)$. All operator norms in this section are the induced norms of this inner product. Retain the exact homogeneous equation

$$U_{xx} - U_{rr} = \mathcal{A}U_x + \mathcal{B}U_r + \mathcal{D}U, \qquad \|\mathcal{A}\|, \|\mathcal{B}\| \le 10^{-4}, \quad \|\mathcal{D}\| \le 10^{-6}. \tag{13}$$

At $r = 0$, the L component is Dirichlet and the four complementary components obey the Robin row

$$U_L = 0, \qquad P_N U_r = M P_N U, \qquad \|M\|_{\mathsf{H}} \le 1/2000, \tag{14}$$

where P_N projects off the L component. The block M need not be self-adjoint or constant. The derivative in the Robin row is the increasing-r derivative in (3), with that sign already included in M. The retained cap domain is $0 \le x \le 3/2$, $\max(0, x - 5/4) \le r \le 7/4 - x$. The cone $\mathcal{K}_T = \{0 \le x \le T,\ 0 \le r \le T - x\}$ lies in the retained coefficient domain; the solution is exactly zero for $x + r \le 1$.

Let $V = U_x$, $P = U_r$, $k = \alpha = 1/2$, and define

$$e = \tfrac{1}{2}(|V|_{\mathsf{H}}^2 + |P|_{\mathsf{H}}^2 + 2k\langle V, P\rangle_{\mathsf{H}} + |U|_{\mathsf{H}}^2),$$
$$\mathcal{E}(x) = \int_0^{T-x} e\,\mathrm{d}r + \tfrac{\alpha}{2}|P_N U(x,0)|_{\mathsf{H}}^2. \tag{15}$$

This is a coercive estimating norm for the linear difference. It is not identified with a positive conserved physical energy of the full shell. Pointwise,

$$e \geq \tfrac{1-k}{2}(|V|_{\mathsf{H}}^2 + |P|_{\mathsf{H}}^2) + \tfrac{1}{2}|U|_{\mathsf{H}}^2. \tag{16}$$

Lemma 1 (Continuous energy and trace bound)**.** *Under* (13)–(14)*, the retained smooth solution satisfies*

$$\mathcal{E}(x) \leq e^{Cx}\mathcal{E}(0), \qquad C = 201/100, \qquad e^{CT} < 13. \tag{17}$$

If $\mathcal{E}(0) \leq R^2/(5\times 10^6)$*, its scalar shell history* $m(x) = U_w(x,0)$ *satisfies*

$$m(x) = 0 \quad (x \leq 1), \qquad |m(x)| \leq R/300 \quad (1 \leq x \leq T). \tag{18}$$

Proof. Put $F = \mathcal{A}V + \mathcal{B}P + \mathcal{D}U$. Direct differentiation gives the local identity

$$e_x = f_r + \langle V + kP, F\rangle_{\mathsf{H}} + \langle U, V\rangle_{\mathsf{H}}, \qquad f = \langle V, P\rangle_{\mathsf{H}} + \tfrac{k}{2}(|V|_{\mathsf{H}}^2 + |P|_{\mathsf{H}}^2). \tag{19}$$

At the shrinking upper endpoint the flux is

$$f - e = -\tfrac{1-k}{2}|V - P|_{\mathsf{H}}^2 - \tfrac{1}{2}|U|_{\mathsf{H}}^2 \leq 0. \tag{20}$$

Thus no boundary data are introduced on that characteristic side.

The Dirichlet row has $U_L = V_L = 0$ at the shell and contributes $-k|P_L|_{\mathsf{H}}^2/2 \leq 0$. Put $u = P_N U(x,0)$ and $v = P_N V(x,0)$. The Robin block and the derivative of the boundary value term give

$$-f_N + \alpha\langle u, v\rangle_{\mathsf{H}} = -\tfrac{k}{2}|v|_{\mathsf{H}}^2 + \langle v, (\alpha I - M)u\rangle_{\mathsf{H}} - \tfrac{k}{2}|Mu|_{\mathsf{H}}^2. \tag{21}$$

This identity does not exchange M with its adjoint. Young's inequality and $\|M\| \leq 1/2000$ control the mixed term using only the operator norm. No derivative of M, Robin quadratic potential, or symmetry assumption is required. More explicitly, put $s = v + Mu$ and $\mu_M = 1/2000$, $\gamma = \alpha + (1-k)\mu_M$. The same shell expression becomes $-k|s|^2/2 - (1-k)\langle s, Mu\rangle_{\mathsf{H}} + (1-k)|Mu|^2 + \alpha\langle u, s - Mu\rangle_{\mathsf{H}}$. Hence it is bounded above by

$$-\tfrac{k}{4}|s|_{\mathsf{H}}^2 + \left\{\frac{\gamma^2}{k} + (1-k)\mu_M^2 + \alpha\mu_M\right\}|u|_{\mathsf{H}}^2. \tag{22}$$

Relative to the boundary summand of $\mathcal{E}$, the resulting rate is $C_{\mathrm{bdry}} = 2003001/1000000 = 2.003001 < C$.

For completeness, if $a_* = b_* = 10^{-4}$ and $d_* = 10^{-6}$, the bulk production is bounded by $c_V|V|^2 + c_P|P|^2 + c_U|U|^2$, where

$$c_V = a_* + (b_* + ka_*)/2 + d_*/2 + 1/2,$$
$$c_P = kb_* + (b_* + ka_*)/2 + kd_*/2, \qquad c_U = d_*/2 + kd_*/2 + 1/2. \tag{23}$$

Consequently its rate relative to the bulk energy is at most $\max\{2c_V/(1-k), 2c_P/(1-k), 2c_U\} = 1000351/500000 = 2.000702 < C$. The shell mixed term in (21) is also absorbed with a rate below C relative to the positive boundary value term. Combining these estimates gives $\mathcal{E}' \leq C\mathcal{E}$ on $0 \leq x < T$,

hence the first inequality in (17). Continuity gives the endpoint statement. A positive Taylor sum and geometric remainder give

$$e^{CT} \le \sum_{j=0}^{24} \frac{(CT)^j}{j!} + \frac{(CT)^{25}/25!}{1 - CT/26} < 13. \tag{24}$$

Finally $|m|^2 \le |P_N U|_{\mathsf{H}}^2 \le 2\mathcal{E}/\alpha$ and

$$\frac{2 \cdot 13}{\alpha(5 \times 10^6)} = \frac{13}{1250000} < \frac{1}{300^2}. \tag{25}$$

Together with the zero wedge this proves (18). □

4 Integrated control of all initial components

The retained continuous background boxes on I give

$$\begin{aligned} 129 \le \rho, \rho_b \le 131, \quad 3/25000 \le -p \le 1/7000, \quad 111/1000 \le a \le 14/125, \\ |p_y| \le 1/4000, \quad |a_y| \le 1/10000, \qquad 1/1200 \le -F_b' \le 1/1000. \end{aligned} \tag{26}$$

Here F_b' is the scalar-field derivative at the background shell of $F = 2\mathcal{V} - \sigma'^2/4 + \sigma^2/3$. Set

$$\begin{aligned} b_+ = \ell\, 131(14/125), \quad g_+ = \ell\, 131(1/7000)^2/[3(111/1000)], \\ H = \Delta(4/3)(2/3)g_+, \qquad H_r = 3\ell^2 H + (b_+ + g_+)H + (2/3)g_+, \qquad V_+ = H_r + b_+ H. \end{aligned} \tag{27}$$

The reconstruction matrix has infinity norm at most $20001/20000$. The retained rational exponential bound on this interval is $4/3$, so variation of constants gives $\sup|h|, \sup|Z| \le H \sup|D|$ and $\sup|h_r| \le H_r \sup|D|$.

On an interval of length $\Delta = 1/4$, the elementary one-dimensional estimate yields

$$\sup|D| \le 2\|D\|_2 + \tfrac{1}{2}\|D_r\|_2 \le \tfrac{\sqrt{17}}{2} R \le \tfrac{21}{10} R. \tag{28}$$

Thus define the per-R L^2 caps

$$\begin{aligned} h_2 &= (21/20)H, & h_{r,2} &= (21/20)H_r, & v_2^0 &= (21/20)V_+, \\ \beta_+ &= 1/1554, & \beta_{r,+} &= \ell\, 131 \left\{ \frac{1/4000}{2(111/1000)} + \frac{(1/7000)(1/10000)}{2(111/1000)^2} \right\}, \\ w_2 &= \beta_+(h_2 + 2/3), & w_{r,2} &= \beta_{r,+}(h_2 + 2/3) + \beta_+(h_{r,2} + 2/3). \end{aligned} \tag{29}$$

For example $\|h\|_2 \le h_2 R$, while both $\|D\|_2$ and $\|D_r\|_2$ are bounded by R in the scalar formulas. Equation (11) gives

$$\|U_0\|_{L^2_{\mathsf{H}}} \le (2h_2 + w_2)R, \quad \|P_0\|_{L^2_{\mathsf{H}}} \le (2h_{r,2} + w_{r,2})R, \quad \|V_0\|_{L^2_{\mathsf{H}}} \le 2v_2^0 R. \tag{30}$$

The initial tilt term is exactly zero: V_0 occupies only the L slot, whereas P_0 has zero L slot. The shell initial value is zero in the unchanged collar. Therefore

$$\mathcal{E}(0) \le c_0 R^2, \quad c_0 = \tfrac{1}{2}\left[(2h_2 + w_2)^2 + (2h_{r,2} + w_{r,2})^2 + (2v_2^0)^2\right] < \frac{1}{5000000}. \tag{31}$$

All these comparisons are rational, including the guard $17/4 < (21/10)^2$. They control all five channels and the complete scalar pullback, rather than only a free scalar history.

5 The specified integrated class and the two proper responses

Use the shell's intrinsic future normal-geodesic foliation, anchored at the common initial section, and set

$$V_\varepsilon(\tau)=\int_{S^3}\mathrm{d}\mu_{h_\varepsilon(\tau)},\qquad \Theta_\varepsilon=\partial_\tau\log V_\varepsilon,\qquad v_2=\frac{\partial_\varepsilon^2 V_\varepsilon|_0}{V_0},\quad \Theta_2=\partial_\varepsilon^2\Theta_\varepsilon|_0. \tag{32}$$

The clock and anchor are transported with a gauge change. The background has $A(\tau)=\rho_b\cosh(\tau/\rho_b)$, $V_0=2\pi^2A^3$, and $\Theta_0=3\rho_b^{-1}\tanh(\tau/\rho_b)>0$ at τ_*.

In the difference from the base, the common first quadratic sources, proper-clock and measure terms cancel. The retained full coupled difference gives

$$\begin{aligned}\nu(T)&:=(v_2-v_2^{\mathrm{base}})(T)=(\rho_b\ell)^2(F_b'/2)\int_1^T K(T,x)m(x)\,\mathrm{d}x,\\ \vartheta(T)&:=(\Theta_2-\Theta_2^{\mathrm{base}})(T)=\rho_b\ell(F_b'/2)\int_1^T K_T(T,x)m(x)\,\mathrm{d}x,\\ K(T,x)&=\frac{\sinh(2\ell(T-x))}{2\ell}\frac{\cosh^2(\ell x)}{\cosh^2(\ell T)},\qquad K_T=\frac{\cosh^2(\ell x)\cosh(\ell T-2\ell x)}{\cosh^3(\ell T)}.\end{aligned} \tag{33}$$

The exact bounds are $0\le K\le T-x$ and $0<K_T\le 1$. The factor $1/2$ in these kernels' source coefficient is retained; there is no extra factor multiplying the differences of the observable second derivatives.

Theorem 1 (Conditional integrated geometric control)**.** *Under the stated retained premises, each admitted smooth compact preparation satisfies*

$$|\nu(T)|\le C_vR,\qquad |\vartheta(T)|\le C_\Theta R,\qquad C_v=\frac{17161}{19200000000000000000},\quad C_\Theta=\frac{131}{2400000000000}. \tag{34}$$

For the specified closed class

$$\mathcal{C}_{H^1}=\{h\ \textit{satisfying}\ (6):\|D\|_{H^1(I)}\le 2\times10^{31}\}, \tag{35}$$

the coefficients at τ_* obey the outward enclosures

$$v_2\in[-7.2400\times10^{16},-8.6400\times10^{15}],\qquad \Theta_2\in[-1.0113\times10^{22},-3.3540\times10^{21}]. \tag{36}$$

For every individual admitted exact family, for sufficiently small nonzero ε of either sign,

$$0<V_\varepsilon(\tau_*)<V_0(\tau_*),\qquad 0<\Theta_\varepsilon(\tau_*)<\Theta_0(\tau_*). \tag{37}$$

The sufficiently small interval can depend on the family.

Proof. Equations (18) and (33) imply

$$C_v=\frac{(131\ell)^2(1/1000)\Delta^2}{4\cdot300},\qquad C_\Theta=\frac{131\ell(1/1000)\Delta}{2\cdot300}, \tag{38}$$

which equal (34). At $R=2\times10^{31}$ the absolute increment caps are below 1.78761×10^{16} and 1.09167×10^{21}. Adding and subtracting the exact caps from the retained base rational intervals, whose outward displays are

$$v_2^{\mathrm{base}}\in[-5.4520,-2.6521]\,10^{16},\qquad \Theta_2^{\mathrm{base}}\in[-9.0208,-4.4458]\,10^{21}, \tag{39}$$

gives (36). The radius and energy/trace gates were declared before evaluation of these final sign gates.

The first integrated volume and expansion variations vanish by the shared quadrupole mean. For each fixed family, retained C^2 dependence gives

$$V_\varepsilon/V_0 = 1 + \tfrac{1}{2}\varepsilon^2 v_2 + o(\varepsilon^2), \qquad \Theta_\varepsilon = \Theta_0 + \tfrac{1}{2}\varepsilon^2\Theta_2 + o(\varepsilon^2). \tag{40}$$

Strict negativity and continuity, with $V_0, \Theta_0 > 0$ at the target, give (37) after a family-dependent shrinkage. □

The estimating norm controls no higher derivatives or nonlinear remainders uniformly. The theorem therefore gives no effective or class-wide nonlinear parameter interval, signed time history, contracting branch, or volume-acceleration sign. Large second coefficients are not finite-amplitude field values.

6 Actual preparation data beyond the old supremum criterion

The earlier sufficient condition used $R_A = \max\{\sup_I |D|, \sup_I |D_r|\} \le 2\times 10^{33}$ and an initial supremum quantity N involving $U_{0,r}$ [7]. A direct H^1 criterion cannot be obtained merely by embedding into that condition. This failure occurs in the actual retained preparation image, not just among arbitrary scalar profiles.

For any smooth Z compactly supported inside I, define $\chi = \ell\rho^3 Z$ and $h = 3bZ - Z_r$. Then $\chi_r = -\ell\rho^3 h$ and $\chi_y = \rho^2 h$, so the exact moment in (6) vanishes by endpoint evaluation. With

$$\mathfrak{A} = \frac{3}{2g}, \qquad c = -2b - g, \qquad d = 3(\ell^2 - b^2 + bg - b_r), \tag{41}$$

elimination of h in (10) gives

$$D = \mathfrak{A}(Z_{rr} + cZ_r + dZ), \quad D_r = \mathfrak{A}Z_{rrr} + (\mathfrak{A}_r + \mathfrak{A}c)Z_{rr} + (\mathfrak{A}_r c + \mathfrak{A}c_r + \mathfrak{A}d)Z_r + (\mathfrak{A}_r d + \mathfrak{A}d_r)Z. \tag{42}$$

The coefficients are bounded on a compact interior interval using $b \in C^2$, $g \in C^1$ and $g > 0$. Literal smooth admissibility uses the retained smooth static background, distinct from the lower C^4 topology for parameter dependence.

Take $r_0 = 9/8$, integers $m \ge 4$, and

$$\psi(u) = \begin{cases} u^3 \exp(1 - 1/(1-u^2)), & |u| < 1, \\ 0, & |u| \ge 1, \end{cases} \qquad Z_m(r) = m^{-5}\psi(m^2(r - r_0)). \tag{43}$$

This is smooth and flat, supported inside $[17/16, 19/16] \subset (1, T)$. The raw metric addition and conditioning generator are consequently compact and flat, with the other canonical sectors still fixed. Change of variables gives the exact scaling

$$\|Z_m^{(j)}\|_{L^2(I)} = m^{2j-6}\|\psi^{(j)}\|_{L^2(-1,1)}, \qquad 0 \le j \le 3. \tag{44}$$

Therefore (42) bounds $\|D_m\|_{H^1(I)}$ by one finite constant $C_* > 0$ independent of m. At the center, $\psi(0) = \psi'(0) = \psi''(0) = 0$ and $\psi'''(0) = 6$, so exactly

$$h_m(r_0) = h_{m,r}(r_0) = D_m(r_0) = 0, \qquad D_{m,r}(r_0) = \frac{9m}{g(r_0)}, \qquad w_{m,r}(r_0) = -\frac{3p(r_0)m}{a(r_0)g(r_0)}. \tag{45}$$

Thus R_A and the old genuine initial derivative supremum, hence every sufficient old N of that form, diverge. The profile amplitudes themselves can tend to zero; the failure concerns derivative suprema.

For any prescribed positive budget B, multiplying every direction by the same $\eta = B/(2C_*) > 0$ puts its H^1 norm below B and preserves this divergence. With $B = 2\times 10^{31}$, infinitely many such data satisfy Theorem 1 while failing the old R_A cutoff. This proves coverage beyond the previous sufficient criterion. It does not state that the new budget ball contains the old one, that the actual responses diverge, or that sign instability has been found. Each fixed m has only its retained family-dependent nonlinear interval.

7 Preparation selection and relation to established methods

The energy proof is a tailored application of standard multiplier methods. Kreiss, Reula, Sarbach and Winicour construct positive mixed-derivative energies, boundary dissipation and lower-order estimates in [1, Sec. 2 and Apps. 1–2]. Their geometric treatment [2] likewise adapts the timelike multiplier to the boundary. Their theorems impose their own boundary conditions; the 2007 displayed condition has a strictly positive time-derivative coefficient. Neither is invoked as a theorem automatically covering the present mixed Dirichlet/nonselfadjoint Robin system or shrinking cone. Equations (19)–(31) provide that direct conditional estimate with the retained model-specific caps.

The estimating energy also does not select the preparation budget. A specified first-tangent quadratic functional $Q(v)$ is common to every preparation here because v is common. Hollands–Wald canonical energy explicitly has such first-tangent dependence in its stationary black-hole setting [3, Sec. 2.3]; that setting and its Killing generator are not supplied by this shell's proper clock. The retained matched-shell Noether calculation supplies necessary quadratic tests of the form

$$DH_K|_0 = 0, \qquad D^2H_K|_0[v,v] = \Omega_{\mathrm{rel},0}(v, \mathcal{K}v) = 0 \tag{46}$$

under its own full-domain and compensated-symmetry premises [8]. Such tests are constant on the admitted second-preparation fiber. They cannot alone select a finite H^1 radius: scaling a fixed nonzero interior profile by any fixed finite parameter changes its R without changing these first-tangent inputs.

The limitation is specific. For an arbitrary C^2 nonlinear functional,

$$(F \circ x_\varepsilon)''|_0 = DF_0[w] + D^2F_0[v,v]. \tag{47}$$

The first term can distinguish preparations. Nor is a bare Noether surface integral automatically a Hamiltonian: generator, boundary conditions, shell and corner completion, and allowed variations matter [4, 5]. A specified local initial-cut functional depending on identical finite boundary and embedding jets has equal coefficients through order two by the chain rule, since these compact additions vanish on the initial collars. This does not apply to arbitrary interior or future cuts, interior-dependent generators, or unspecified physical energies. No no-go theorem for every energy or state-based preparation rule is claimed.

The physical initial section, continued component, scalar-label calibration, canonical-sector restriction and numerical radius remain inputs. The present result certifies what follows from one integrated geometric specification; it supplies no action, state or process that chooses it. No measured amplitude, matter/radiation energy transfer or higher-dimensional Big Bang origin is established. The focused primary comparison identifies the standard methods used here, without asserting exhaustive novelty or priority.

A Certificates and reproducibility

The companion capsule [9] contains the tilted-energy proof, predeclared design, selected-source SHA-256 pins, exact rational producer, normal and optimized reports, independent reconstruction, actual-image concentration proof and its algebra checker, and the limited preparation-selection and primary-source comparisons. The producer authenticates retained records before checking rational consequences. Authentication of a published coefficient assertion is not a fresh proof of that continuous assertion.

The 24 finite arithmetic gates verify the positive tilt, bulk and nonsymmetric Robin rate guards, exponential remainder, complete initial map, boundary trace factor, and whole-class sign displays. The concentration checker verifies operator identities, center jets, support exponents and continuous analytic bump bounds without evaluating a numerical bump. No new background/evolved-field oracle, PDE solve or numerical bump query is used. Independent implementations share the retained analytic

and background premises; their agreement is not an independent observation of the nonlinear PDE. Source, displayed-bound audit, deterministic compilation receipt and page-render review accompany this working draft.

References

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